Every Other One — Loops + If

AP CSP · Unit 3, Lesson 5 · Reference Sheet
Read page 1 before class

Put an `if` inside a loop and every trip can decide for itself: which color, or whether to draw at all. With `%` and `!=` you get every other one, every third one, checkerboards and gaps.

An if inside the loop

LOOP + IF

```
for (let i = 0; i < 14; i++) {
  if (i % 2 == 0) {
    fill(230, 60, 60);    // even: red
  } else {
    fill(60, 190, 90);    // odd: green
  }
  circle(80 + i * 50, 300, 30);
}
```

EVERY TRIP
The loop decides **how many**. The if decides **what each one looks like**. 14 trips, 14 decisions.

The patterns

% AND !=

CONDITION	TRUE FOR I =
`i % 2 == 0`	0, 2, 4, 6 …
`i % 2 == 1`	1, 3, 5, 7 …
`i % 3 == 0`	0, 3, 6, 9 …
`i != 4`	every one except 4
`i % 2 == 0 && i != 4`	0, 2, 6, 8 …

EVERY THIRD STARTS AT 0
`i % 3 == 0` picks the 1st, 4th, 7th … one, because i starts at 0.

Checkerboards

(r + c) % 2

```
if ((r + c) % 2 == 0) { fill(240); }
else { fill(120, 160, 220); }
rect(c * 100, r * 100, 100, 100);
```

CONDITION	PATTERN
`(r + c) % 2`	checkerboard
`c % 2`	vertical stripes
`r % 2`	horizontal stripes

Special case first

ELSE IF

```
if (i == RIVER) { water }        // 4
else if (i % 2 == 1) { grass }
else { road }
```

FIRST TRUE ONE WINS
Exactly one branch runs. RIVER is 4, an even row; if the road check came first it would take row 4. Put the **most specific** condition first.

On the AP exam

REFERENCE SHEET

```
i ← 0
REPEAT 14 TIMES
{
  IF (i MOD 2 = 0)
  { count ← count + 1 }
  i ← i + 1
}
```

AAP-2.H · AAP-2.K
MOD is `%`. In pseudocode `=` **compares** (like `==`) and `←` **assigns** (like `=`). This one counts the evens: 7.

Check yourself before class

1. i from 0 to 9: how many make `i % 3 == 0`?
2. Condition that skips only i = 7?
3. `(r + c) % 2 == 0` at r = 2, c = 3?
4. Which i from 0 to 9 pass `i % 2 == 1 && i != 5`?
5. What does `c % 2` alone draw on a grid?
6. In pseudocode, what does `=` mean?

Answers. 1) 4 (0, 3, 6, 9). 2) `i != 7`. 3) False (5 % 2 = 1). 4) 1, 3, 7, 9. 5) Vertical stripes. 6) Compare (is equal to).

PAGE 2 — after the build. The patterns behind TODOs 1–5 and the labs.

A loop inside an if inside a loop — TODO 2

```
for (let i = 0; i < ROWS; i++) {
  if (i == RIVER) { … }
  else if (i % 2 == 1) {
    for (let c = 0; c < 16; c++) {
      if ((i + c) % 2 == 0) { fill(120, 200, 100); }
      else { fill(95, 170, 80); }
      rect(c * TILE, i * ROW_H, TILE, ROW_H);
    }
  }
  else { … road … }
}
```

Only grass rows need tiles, so the tile loop lives inside the grass branch. Line up your braces.

Change the color, or skip it? — TODO 4

```
if (c != 4) {                 // skip: nothing at a
  if (c % 3 == 0) {           // decide the color
    fill(230, 50, 50);
    red = red + 1;
  } else {
    fill(240);
  }
  rect(x, y, 70, 32, 6);
  cars = cars + 1;
}
```

SKIP MEANS EVERYTHING

The skip has to wrap the color, the rect **and** the counters. Wra only the rect and the readout counts a car you cannot see (32 instead of 28).

Two conditions at once — &&

`i % 2 == 0 && i != RIVER` — even **and** not the river. Both must be true. From Unit 2.

Grey or gone?

`if (i == 6) { fill(60); }` draws bulb 6 grey. `if (i != 6) { … circle(…); }` does not draw it at all.

Three mistakes worth knowing about in advance

WRITTEN LIKE THIS	WHAT ACTUALLY HAPPENS
`if (i % 2 = 0)`	`=` assigns. Use `==` to compare.
`if (i % 2 == 1) {grass} else if (i == RIVER)`	Fine here (4 is even) — but put special cases first anyway, or the next change breaks it.
`if (c % 2 == 0)` *for a checkerboard*	Stripes. A checkerboard needs both: `(r + c) % 2`.

AP CSP 2026–27 · Unit 3 Lesson 5 · 3.8 Iteration (AAP-2.K) · 3.6 Conditionals (AAP-2.H) · Skills 2.B · 4.B